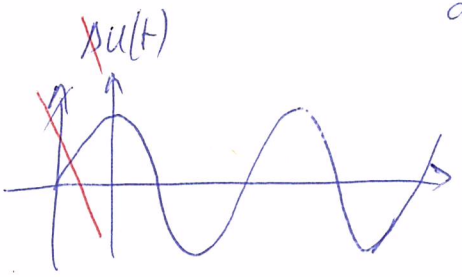
12

Alternating current (AC) circuits

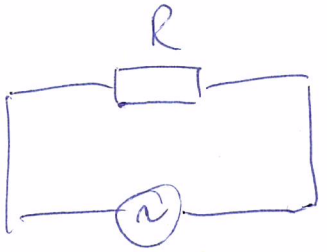
An AC source or power supply provides alternating voltage:

$$\Delta U(t) = \Delta U_{max} \cdot \cos \omega t$$

ΔU_{max}: amplitude; ω: angular frequency



Typical AC source is a generator, that is a coil rotating at ω in a magnetic field.



$$\Delta U(t) - I(t)R = 0 \quad \text{(Kirchhoff II.)}$$

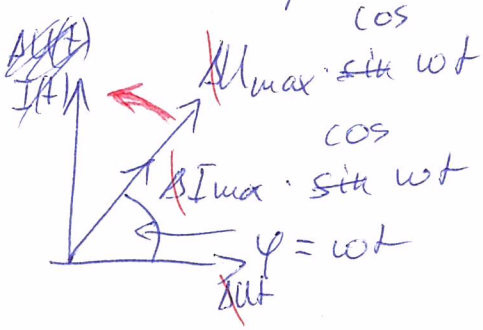
$$\Downarrow$$

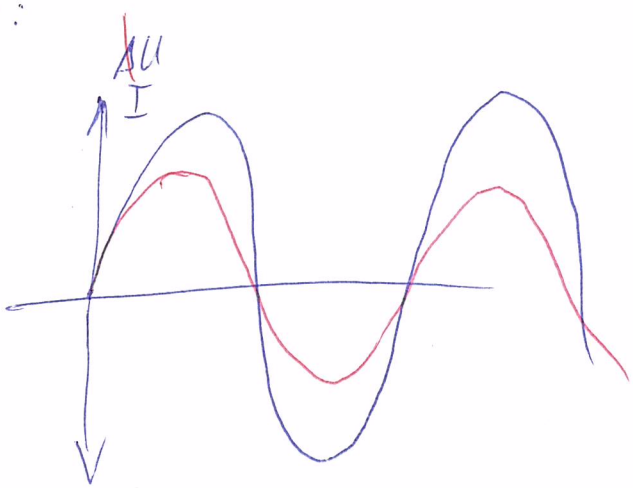
$$I(t) = \frac{\Delta U(t)}{R} = \frac{\Delta U_{max}}{R} \cdot \cos \omega t$$

$$\frac{\Delta U_{max}}{R} = I_{max}$$

For a resistive load $\Delta U(t)$ and $I(t)$ are in phase

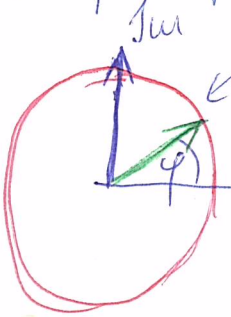
Phasor representation:





Complex notation:

Complex phasor



complex unit-radius circle. Take a unit vector $\underset{\text{unit}}{u}$

$$u(\varphi) = \mathrm{Re}\,u + i\,\mathrm{Im}\,u = \cos(\varphi) + i\sin(\varphi)$$

Rotating vector: $e^{i\varphi} = e^{i\omega t}$

$$V(t) = V_0 \cdot e^{i\omega t}$$

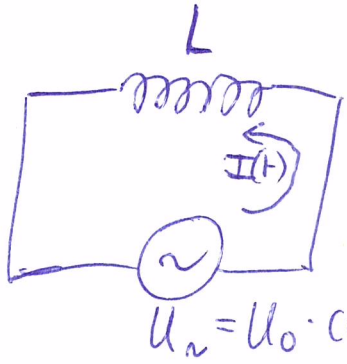
physical quantity (voltage, current, displacement, speed)

$$\cos\alpha = \frac{e^{i\alpha} + e^{-i\alpha}}{2}$$

$$\sin\alpha = \frac{e^{i\alpha} - e^{-i\alpha}}{2i}$$

$$V(t) = V_0 \cdot \cos\omega t = \mathrm{Re}\, V_0 e^{i\omega t}$$

Inductor in an AC circuit



$$U_N - L\frac{dI}{dt} = 0$$

$$U_0 \cos\omega t = L\frac{dI}{dt}$$

$$dI = \frac{U_0}{L}\cos\omega t \cdot dt$$

$$I(t) = \frac{U_0}{L}\int_0^t \cos(\omega t')\,dt' = \frac{U_0}{L\omega}\cdot\sin(\omega t) =$$

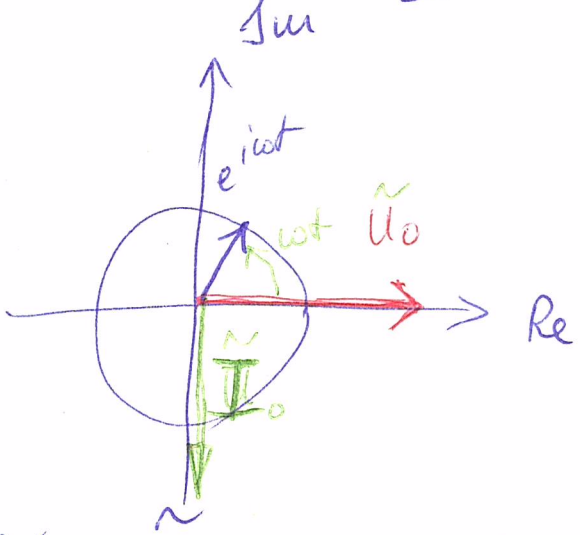
$$I(t) = \frac{U_0}{L\omega}\cos\left(\omega t - \frac{\pi}{2}\right) \qquad I_0 = \frac{U_0}{L\omega}$$ (Ohm like)

$$\varphi_I = \varphi_U - \frac{\pi}{2}$$

The phase of the current lags behind that of voltage

In complex notation:

$$U_n = \mathrm{Re}\left[\tilde{U}_0 e^{i\omega t}\right] \qquad \tilde{U}_0 e^{i\omega t} = L\frac{dI}{dt}$$



$$dI = \frac{\tilde{U}_0}{L} e^{i\omega t} dt$$

$$I(t) = \frac{\tilde{U}_0}{L} \cdot \int_0^t e^{i\omega t'} dt' =$$

$$= \underbrace{\frac{\tilde{U}_0}{i\omega L}} \cdot e^{i\omega t}$$

$$\tilde{I}_0 = \frac{\tilde{U}_0}{i\omega L} = \frac{-i\tilde{U}}{\omega L}$$

$\tilde{I}_0$ complex amplitude of I

The $\tilde{I}$ lagging behind $\tilde{U}_0$ is very apparent!

$$I_0 = \frac{U_0}{L\omega}$$ ← resembles to Resistance

$$X_L = \omega L$$

inductive reactance or impedance

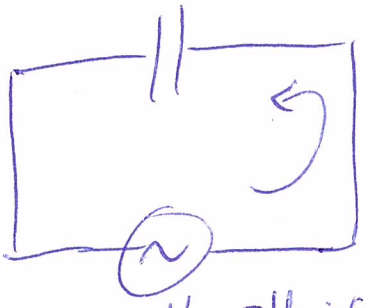
$$\tilde{I}_0 = \frac{\tilde{U}_0}{i\omega L}$$

$$\tilde{X}_L = i\omega L$$

complex impedance

$$I = \frac{U_0}{L\omega}\cos\left(\omega t + \frac{\pi}{2}\right)$$

Capacitor in an AC-circuit



$U_N = U_0 \cdot \cos\omega t = \mathrm{Re}\left(U_0 e^{i\omega t}\right)$

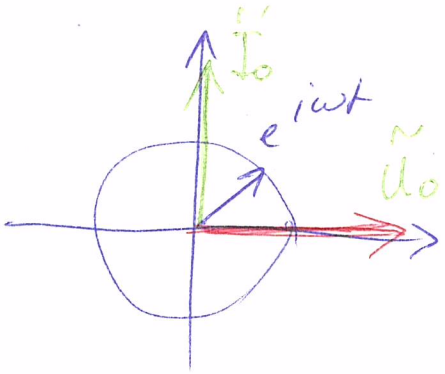
$U_0 e^{i\omega t} + \frac{Q(t)}{C} = 0$

$\hat{Q}(t) = C \tilde{U}_0 e^{i\omega t}$

$I = \frac{dQ}{dt} = +i\omega C \tilde{U}_0 e^{i\omega t}$

$\tilde{I}(t) = +i\omega C \tilde{U}_0 e^{i\omega t} = \frac{\tilde{U}(t)}{1/i\omega C}$

$$X_C = \frac{1}{i\omega C}$$



I is ahead of U in a capacitor!

Forced oscillation for RLC in complex phasors!

(Ch 19?) I is const. (connected in series) → phasors for $\hat{U}$

Power in an AC circuit

For DC: the power of the battery was simply $U \cdot I$

Here also the instantaneous power is $U(t) \cdot I(t)$

$$P(t) = U_0 \cos(\omega t) \cdot I_0 \cos(\omega t + \varphi_0)$$

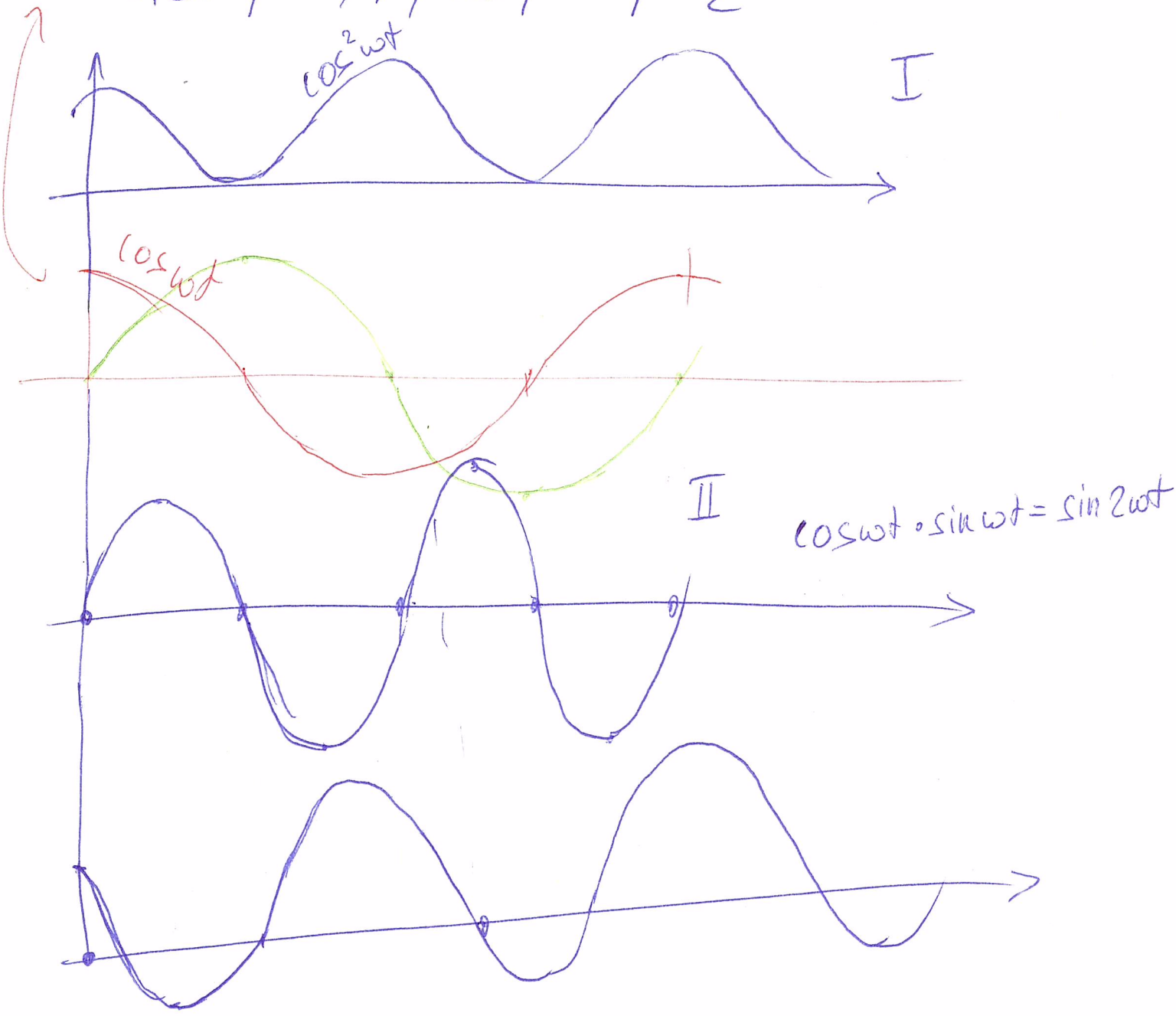
I

$\cos\omega t \cdot \cos\varphi_0 - \sin\omega t \cdot \sin\varphi_0$

$$P = U_0 I_0 \cdot \left[\cos^2\omega t \cdot \cos\varphi_0 - \cos(\omega t) \cdot \sin(\omega t) \cdot \sin\varphi_0\right]$$

quite complex function of time in general

E.G. $\varphi = \pi/4$, $\cos\varphi = \sin\varphi = \frac{\sqrt{2}}{2}$



What we are interested in is the time average of the power for a single periode (or more.)

time average of $\cos^2(\omega t) = \frac{1}{2}$ I

$\cos(\omega t) \cdot \sin(\omega t) = \emptyset$ II

so the time average of the II nd term is $\emptyset$

$$P_{avg} = \frac{1}{2} U_0 I_0 \cdot \cos\varphi$$

- if the load is purely resistive $\varphi = 0$
  full $\frac{1}{2} U_0 I_0$ power is pulled out from power supply
- if the load is purely capacitive or inductive
  $\varphi = \pm \frac{\pi}{2}$, $\emptyset$ power is consumed!
  Energy is stored in C and L and returned to the circuit !!!
- For a given R, $X_C = X_L$ provides a $\emptyset$ imaginary part of the complex impedance,
  ↳ maximum power can be used.

For having the same old expression of $P = UI$ it is customary to introduce $U_{RMS} = \frac{U_0}{\sqrt{2}}$

$I_{RMS} = \frac{I_0}{\sqrt{2}}$ effective values, for which

$$P = U_{RMS} \cdot I_{RMS} = I_{RMS}^2 \cdot R$$

So in the home 230V $\overset{\text{power network}}{\rightarrow}$ $U_{RMS} = 230V$!
$U_0 = 325V$

Note: complex notation is more difficult
$P(t) = Re(\tilde{U}(t)) \cdot Re(\tilde{I}(t))$ because $Re(\tilde{U} \cdot \tilde{I}) \neq Re\tilde{U} \cdot Re\tilde{I}$

cos

$$P = Re(\tilde{U}) \cdot Re(\tilde{I}) = U_0 \cdot \frac{e^{i\omega t} + e^{-i\omega t}}{2} \cdot I_0 \frac{e^{i(\omega t+\varphi)} + e^{-i(\omega t+\varphi)}}{2} = \quad (F)$$

$$= \frac{U_0 I_0}{4}\left(e^{i2\omega t+\varphi} + e^{-2\omega t-\varphi} + e^{i\varphi} + e^{-i\varphi}\right) =$$

$$= \frac{U_0 I_0}{2}\left[\cos(2\omega t+\varphi) + \cos\varphi\right]$$

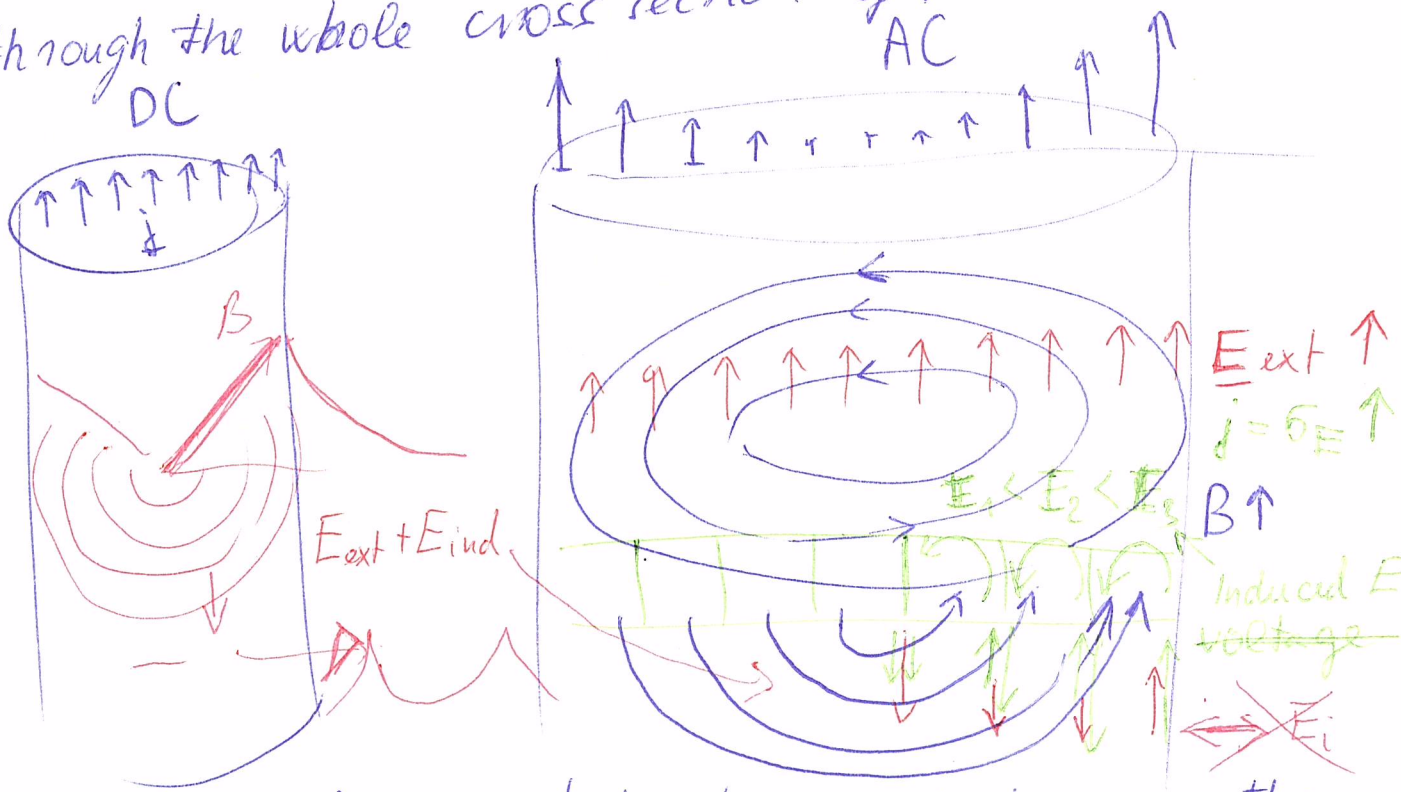
$$P_{avg} = \frac{U_0 I_0}{2}\cos\varphi$$

Simpler: $\tilde{P}_{avg} = \frac{U \cdot I^*}{2} = U_0 e^{i\omega t} \cdot I_0 \cdot e^{-i(\omega t+\varphi)} =$

$= \frac{U_0 I_0}{2} \cdot e^{-i\varphi}$ $\quad Re\,\tilde{P}_{avg} = \frac{U_0 I_0}{2}\cos\varphi$

Skin effect

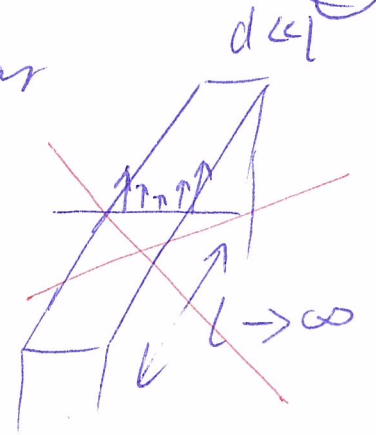
In a DC current the current density is constant through the whole cross section of the conductor



while in AC current density concentrates on the outside of the conductor Why?

Quantitatively :

Imagine our conductor to be rectangular



① Ohm's law $\underline{j} = \sigma \underline{E}$

② Ampère's law : $\operatorname{rot} \underline{H} = \underline{j}$

③ Faraday's law $\operatorname{rot} \underline{E}_{ind} = -\frac{d\underline{B}}{dt}$

rot

$\operatorname{rot} \underline{B} = \mu \underline{j} \quad / \frac{\partial}{\partial t}$

$\operatorname{rot} \operatorname{rot} E_{ind} = -\operatorname{rot} \frac{\partial \underline{B}}{\partial t}$

$\operatorname{rot} \frac{\partial \underline{B}}{\partial t} = \mu_0 \frac{\partial \underline{j}}{\partial t}$

$\operatorname{rot}(\operatorname{rot} E_i) = -\mu_0 \frac{\partial \underline{j}}{\partial t}$

$\underbrace{\operatorname{grad}(\operatorname{div} E_i)}_{0 \text{ since } \frac{\varrho}{\varepsilon_0} = 0} - \Delta \underline{E}_i = -\mu \frac{\partial \underline{j}}{\partial t}$

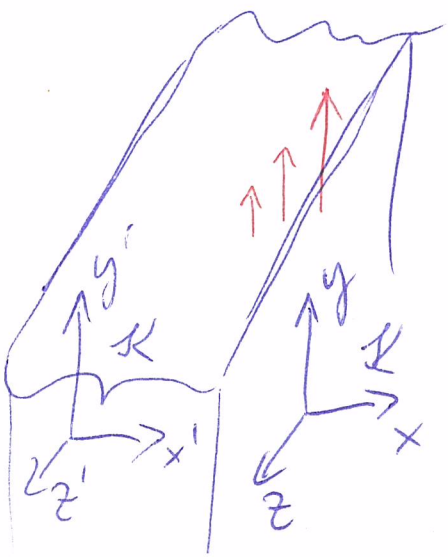
$\underline{E} = \frac{\underline{j}}{\sigma}$

$\Delta \underline{j} = \mu \sigma \frac{\partial \underline{j}}{\partial t}$

Test function :

$\underline{j}(\underline{r}, t) = \underline{j}_0(\underline{r}) \cdot e^{i\omega t}$

$\Delta \underline{j}_0 = i\omega \sigma \mu \cdot \underline{j}_0(r)$



We are interested in the y component

x dependence of the

y and z derivatives $= 0$ $\left(\Delta \underline{V} = \frac{\partial^2 \underline{V}}{\partial x^2} + \frac{\partial^2 \underline{V}}{\partial y^2} + \frac{\partial^2 \underline{V}}{\partial z^2} \right)$

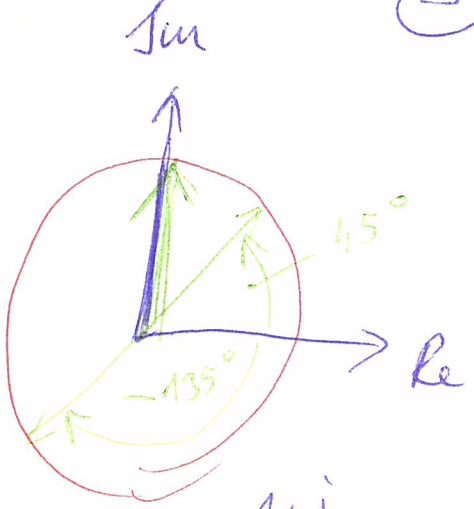
$\frac{\partial^2 j_{0y}}{\partial x^2} = i\omega \sigma \mu \, j_{0y} \quad / \; j_{0y} := A e^{\lambda x}$

$$A \cdot \lambda^2 \cdot e^{\lambda x} = i\omega\sigma\mu A e^{\lambda x}$$

$$\lambda^2 = i\omega\sigma\mu =$$

$$\lambda_{1,2} = \sqrt{i} \cdot \sqrt{\omega\sigma\mu} = \cancel{\quad} =$$

$$= \pm \frac{1+i}{\sqrt{2}} \sqrt{\omega\sigma\mu}$$

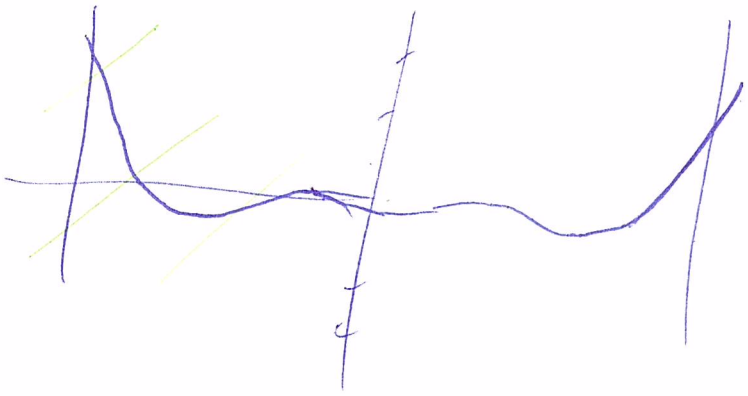


$$\pm \frac{1+i}{\sqrt{2}}$$

$$\delta := \sqrt{\frac{2}{\omega\sigma\mu}}$$

$$\left[\left(\frac{1+i}{\sqrt{2}} \right)^2 = \frac{1+2i-1}{2} = i \right]$$

$$j_{0y} = \mathrm{Re}\, A \left[\underbrace{e^{-\frac{x}{\delta}} \cdot e^{-i\frac{x}{\delta}}}_{\text{in } \mathbb{K}'} + \underbrace{e^{\frac{x}{\delta}} \cdot e^{+i\frac{x}{\delta}}}_{\text{physical solution in } \mathbb{K}} \right]$$



δ : skin depth

$\frac{\omega}{2\pi} = 50\,\mathrm{Hz}$ $\delta = 8\,\mathrm{mm}$

$10\,\mathrm{MHz}$ $\delta = 21\,\mu\mathrm{m}$

$10\,\mathrm{GHz}$ $\delta = 0{,}65\,\mu\mathrm{m}$

Ch 20 : Calculate skin eff in cylindrical Coordinates System !